ChIP-seq Analysis & Peak Calling Lab Report

Name: Debarshi Das Adhikary

Roll No.: 255hsbb011

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1. Aim

To perform end-to-end processing of paired-end ChIP-seq data for Wild Type (WT) and Mutant samples up to peak calling, generate quality-controlled alignments and both narrow/broad peak profiles against matching Input controls, and visualize the signal enrichment in the Integrative Genomics Viewer (IGV) in preparation for downstream differential binding analysis (DiffBind).

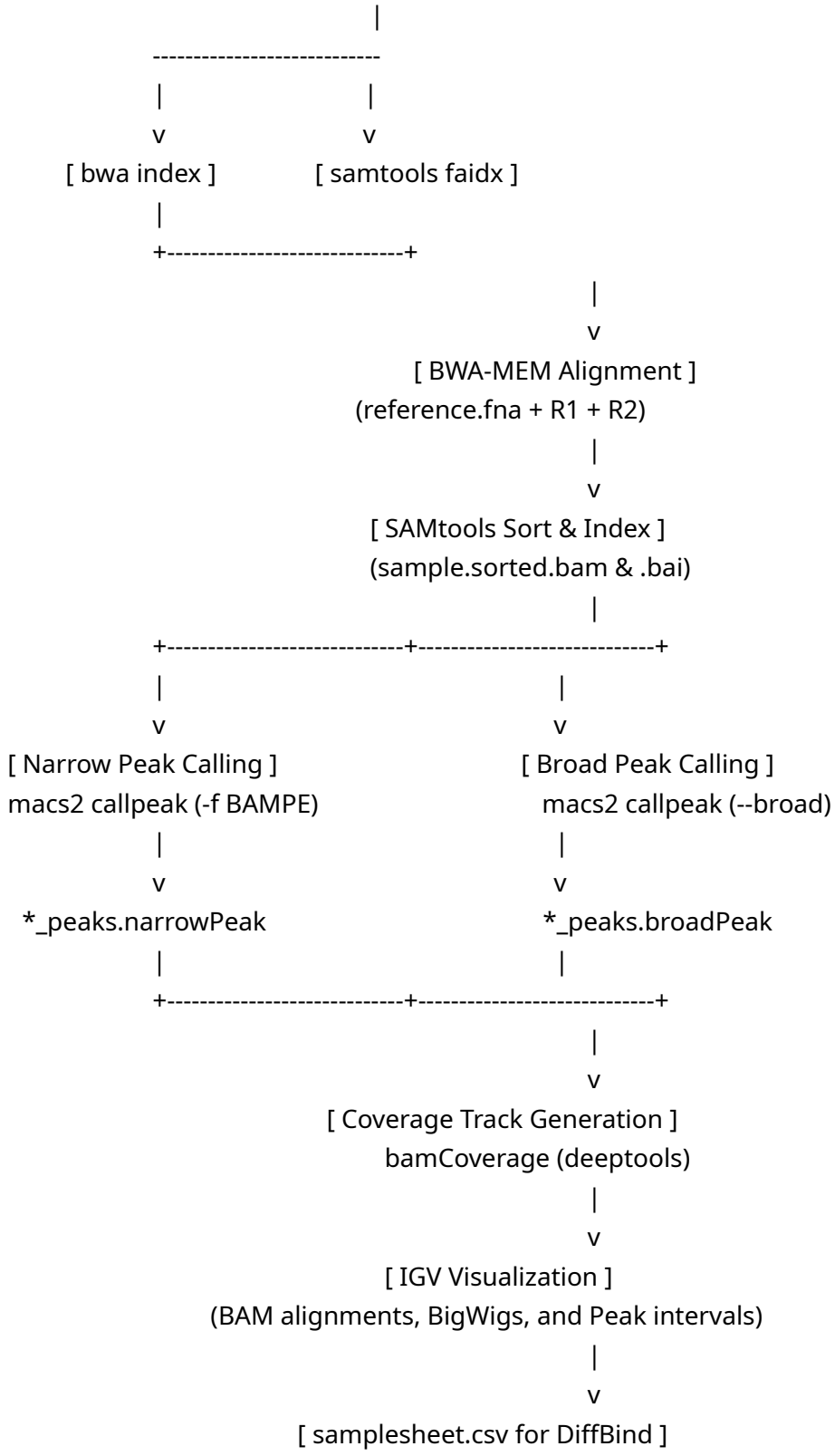
2. Experimental Design & Dataset Summary

Condition	Biological Replicate	ChIP Sample (Treatment)	Input Sample (Control)
Wild Type (WT)	Replicate 1	C1_R1 & C1_R2	I1_R1 & I1_R2
	Replicate 2	C2_R1 & C2_R2	I2_R1 & I2_R2
Mutant	Replicate 1	MC1_R1 & MC1_R2	MI1_R1 & MI1_R2
	Replicate 2	MC2_R1 & MC2_R2	MI2_R1 & MI2_R2

- **Sequencing Type:** Paired-End (PE) Illumina sequencing (_R1 and _R2).
- **Reference Genome:** reference.fna (shared across both WT and Mutant strains).

3. Workflow Flowchart

[Paired-End FASTQ Files: Treatment & Input]
|
v
[Reference Genome: reference.fna]



4. Materials and Computational Environment Setup

Bioconda packages are loaded into the Pixi environment:

Bash

```
# Add bioconda channel to project configuration
pixi project channel add bioconda
```

```
# Install required bioinformatics tools
pixi add bwa samtools macs2 deeptools
```

Tool Descriptions:

- bwa **(Burrows-Wheeler Aligner):** Maps low-divergent sequencing reads against large reference genomes.

- samtools**:** Manipulates, sorts, filters, and indexes high-throughput sequencing alignments.

- macs2 **(Model-based Analysis of ChIP-Seq):** Identifies transcription factor binding sites and histone modification domains from ChIP-seq data.

- deeptools **(**bamCoverage**):** Computes normalized continuous read coverage profiles (.bigWig) across the entire genome.

5. Execution Steps & Commands

All operations are executed in the working directory containing reference.fna and the sample FASTQ pairs.

Step 1: Reference Genome Indexing

Indexing creates fast-lookup Burrows-Wheeler Transform (BWT) tables and coordinate fasta indices (.fai) required for read mapping and coordinate extraction.

Bash

```
bwa index reference.fna
samtools faidx reference.fna
```

Step 2: Read Alignment, Coordinate Sorting, and Indexing

Each paired-end dataset is aligned to the indexed reference using bwa mem, then piped directly into samtools sort to generate coordinate-sorted BAM files, followed by index creation (.bai).

A. Mutant Replicate 1 (ChIP & Input):

Bash

```
# ChIP treatment
bwa mem -t 10 reference.fna MC1_R1.fastq.gz MC1_R2.fastq.gz | samtools sort -@ 4 -m 2G -o MC1.sorted.bam && samtools index -@ 4 MC1.sorted.bam

# Input control
```

bwa mem -t 10 reference.fna MI1_R1.fastq.gz MI1_R2.fastq.gz | samtools sort -@ 4 -m 2G -o MI1.sorted.bam && samtools index -@ 4 MI1.sorted.bam

B. Mutant Replicate 2 (ChIP & Input):

Bash

```
# ChIP treatment
bwa mem -t 10 reference.fna MC2_R1.fastq.gz MC2_R2.fastq.gz | samtools sort -@ 4 -m 2G -o MC2.sorted.bam && samtools index -@ 4 MC2.sorted.bam

# Input control
bwa mem -t 10 reference.fna MI2_R1.fastq.gz MI2_R2.fastq.gz | samtools sort -@ 4 -m 2G -o MI2.sorted.bam && samtools index -@ 4 MI2.sorted.bam
```

C. Wild Type Replicates 1 & 2 (ChIP & Input):

Bash

```
# WT Replicate 1
bwa mem -t 10 reference.fna C1_R1.fastq.gz C1_R2.fastq.gz | samtools sort -@ 4 -m 2G -o C1.sorted.bam && samtools index -@ 4 C1.sorted.bam

#Input control
bwa mem -t 10 reference.fna I1_R1.fastq.gz I1_R2.fastq.gz | samtools sort -@ 4 -m 2G -o I1.sorted.bam && samtools index -@ 4 I1.sorted.bam

# WT Replicate 2
bwa mem -t 10 reference.fna C2_R1.fastq.gz C2_R2.fastq.gz | samtools sort -@ 4 -m 2G -o C2.sorted.bam && samtools index -@ 4 C2.sorted.bam
```

Input control

bwa mem -t 10 reference.fna I2_R1.fastq.gz I2_R2.fastq.gz | samtools sort -@ 4 -m 2G -o I2.sorted.bam && samtools index -@ 4 I2.sorted.bam

Step 3: Peak Calling with MACS2

Peak calling identifies statistically significant loci enriched in the ChIP sample relative to the local non-specific background captured by the Input control.

- Flag -f BAMPE instructs MACS2 to use the actual insert sizes derived from paired-end alignments.
- Flag -B exports bedGraph signal tracks for both ChIP and background.
- Replace -g hs with the appropriate effective genome size if working with a non-human model (e.g., -g mm for mouse, -g 1.2e7 for yeast).

3.1. Narrow Peak Calling (Punctate Binding / Transcription Factors)

Bash

```
# Mutant Replicate 1
# Mutant Replicates
macs3 callpeak -t MC1.sorted.bam -c MI1.sorted.bam -f BAMPE -g hs -n Mutant_Rep1_narrow -B
macs3 callpeak -t MC2.sorted.bam -c MI2.sorted.bam -f BAMPE -g hs -n Mutant_Rep2_narrow -B

# WT Replicates
macs3 callpeak -t C1.sorted.bam -c I1.sorted.bam -f BAMPE -g hs -n WT_Rep1_narrow -B
macs3 callpeak -t C2.sorted.bam -c I2.sorted.bam -f BAMPE -g hs -n WT_Rep2_narrow -B
```

3.2. Broad Peak Calling (Diffuse Enrichment / Histone Marks)

Bash

```
# Mutant Replicates
macs3 callpeak -t MC1.sorted.bam -c MI1.sorted.bam -f BAMPE -g hs --broad -n
Mutant_Rep1_broad -B
macs3 callpeak -t MC2.sorted.bam -c MI2.sorted.bam -f BAMPE -g hs --broad -n
Mutant_Rep2_broad -B

# WT Replicates
macs3 callpeak -t C1.sorted.bam -c I1.sorted.bam -f BAMPE -g hs --broad -n WT_Rep1_broad -B
macs3 callpeak -t C2.sorted.bam -c I2.sorted.bam -f BAMPE -g hs --broad -n WT_Rep2_broad -B
```

Step 4: BigWig Track Generation (Normalized Signal Tracks)

To facilitate direct visual comparisons of binding intensities between samples regardless of differences in sequencing depth, BAM alignments are converted to RPKM-normalized BigWig tracks:

Bash

```
bamCoverage -b MC1.sorted.bam -o MC1.rpkm.bw --normalizeUsing RPKM --binSize 10 -p 12
bamCoverage -b MC2.sorted.bam -o MC2.rpkm.bw --normalizeUsing RPKM --binSize 10 -p 12
bamCoverage -b C1.sorted.bam -o C1.rpkm.bw --normalizeUsing RPKM --binSize 10 -p 12
bamCoverage -b C2.sorted.bam -o C2.rpkm.bw --normalizeUsing RPKM --binSize 10 -p 12
```

Step 5: IGV (Integrative Genomics Viewer) Visualization

1. Launch IGV.

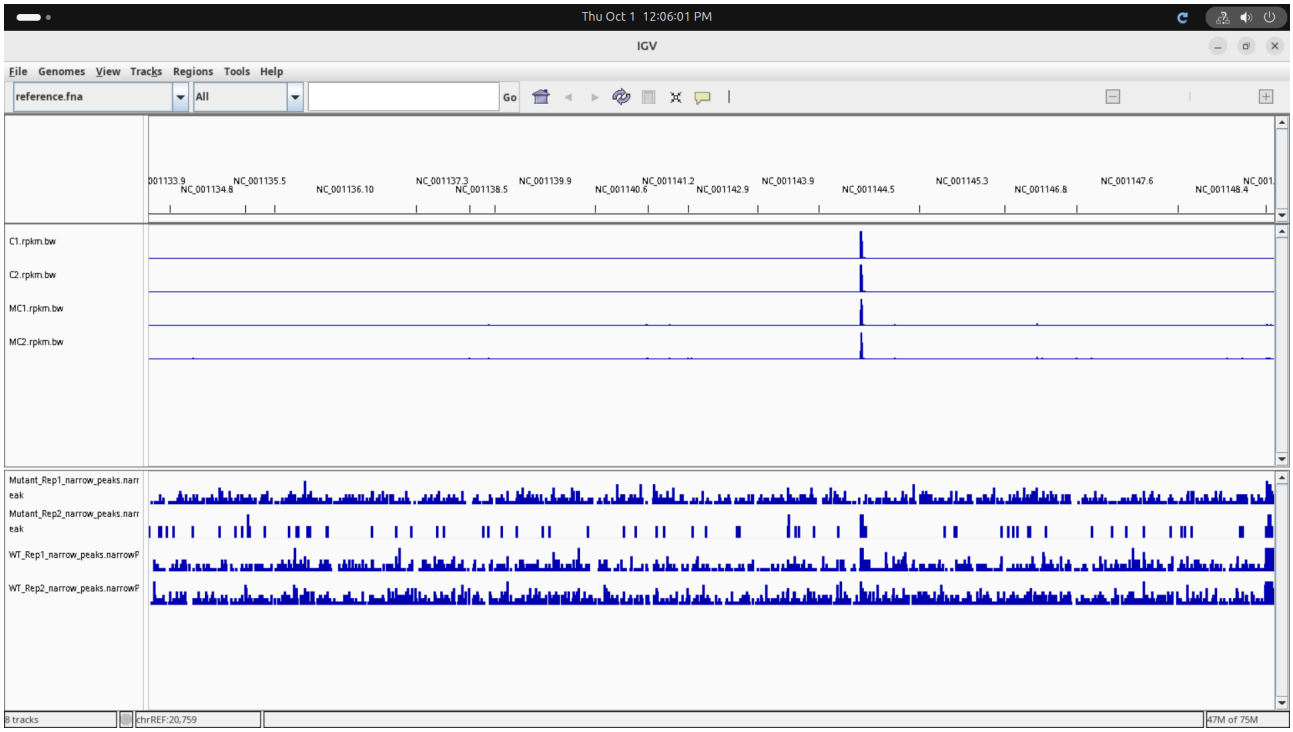
2. Select **Genomes → Load Genome from File...** and select reference.fna.

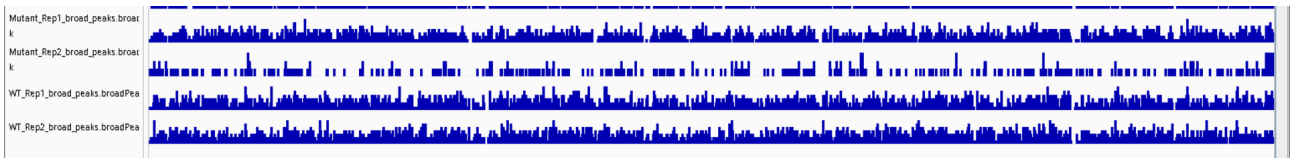
3. Select **File → Load from File...** and load:

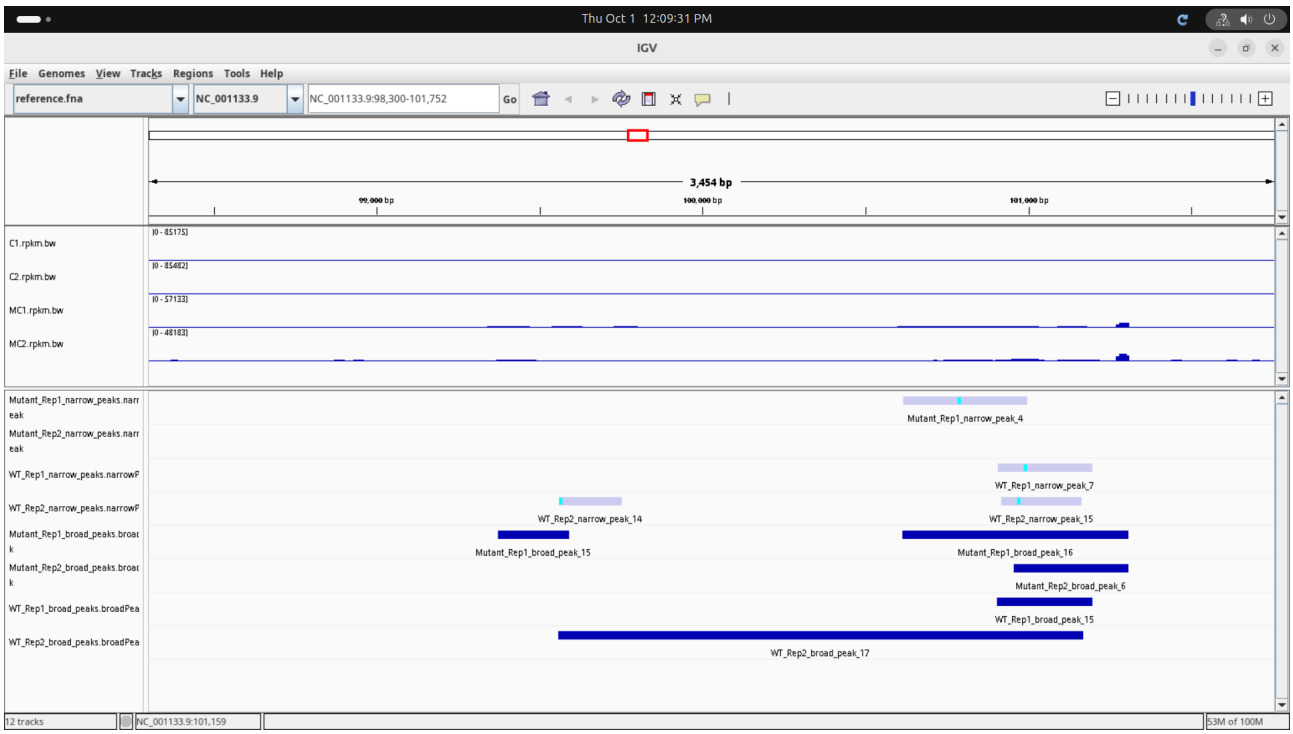
- Coverage tracks: C1.rpkm.bw, C2.rpkm.bw, MC1.rpkm.bw, MC2.rpkm.bw

- Peak intervals: *_peaks.narrowPeak and *_peaks.broadPeak

4. Set Track Scaling: Highlight all coverage tracks, right-click, and select **Set Data Range** to use an identical scale (e.g., 0 - 50) across both WT and Mutant tracks to prevent misinterpreting baseline differences as true biological signal variations.







6. Output Files & Results Interpretation

1. Alignment Artifacts (.bam & .bai)

- **BAM (Binary Alignment Map):** Stores compressed, coordinate-sorted reads mapped against the reference genome.

- **BAI Index:** Allows fast random-access retrieval of read alignments for specific genomic windows in IGV.

2. MACS2 Output Files

- *_peaks.narrowPeak **(BED 6+4 format):** Loci showing point-source binding enrichment.

- *Column 5:* Normalized score (-10 * log10(q-value)).

- *Column 7:* Signal value (fold enrichment).

- *Column 10:* Peak summit coordinate relative to chromStart (the single base position with the highest read coverage).

- *_peaks.broadPeak **(BED 6+3 format):** Loci showing diffuse binding domains without an explicit single-base summit.

- *_peaks.xls**:** Tab-delimited text summary containing peak coordinates, pileup values, fold enrichment, and significance statistics ($-\log_{10} p$-value and $-\log_{10} q$-value).

3. Biological & Analytical Interpretation

- **Peak Concordance between Replicates:** High-confidence binding events should be present in both biological replicates (e.g., overlapping peak coordinates between WT_Rep1 and WT_Rep2). Discrepant peaks often represent technical noise, PCR amplification bias, or low read depth.

- **ChIP Signal vs. Input Control:** True protein-DNA interaction sites show sharp enrichment in ChIP samples accompanied by low, flat coverage in matching Input controls. Genomic regions with elevated signal in both ChIP and Input represent non-specific binding artifacts, copy-number amplifications, or unannotated repetitive regions.

- **Narrow vs. Broad Geometry:**

- Sharp, tall peaks with high fold enrichment are characteristic of sequence-specific transcription factor binding sites.

- Wide, plateau-like peaks spanning multiple kilobases represent chromatin-modifying complexes or histone post-translational modifications (e.g., H3K27me3, H3K9me3).

7. Preparation for Differential Binding Analysis (DiffBind)

To prepare for downstream quantitative comparison of peak affinities between WT and Mutant conditions in R using DiffBind, generate the metadata sheet samplesheet.csv:

Code snippet

```
SampleID,Condition,Replicate,bamReads,bamControl,Peaks,PeakCaller
WT_R1,WT,1,C1.sorted.bam,I1.sorted.bam,WT_Rep1_narrow_peaks.narrowPeak,narrow
WT_R2,WT,2,C2.sorted.bam,I2.sorted.bam,WT_Rep2_narrow_peaks.narrowPeak,narrow
Mutant_R1,Mutant,1,MC1.sorted.bam,MI1.sorted.bam,Mutant_Rep1_narrow_peaks.narrowPeak,narrow
```

Mutant_R2,Mutant,2,MC2.sorted.bam,MI2.sorted.bam,Mutant_Rep2_narrow_peaks.narrowPeak,narrow

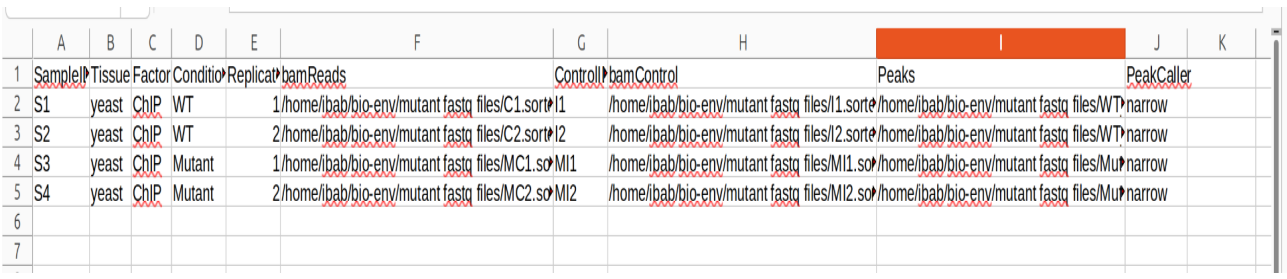
This table connects each condition, replicate, aligned reads file, matching control, and peak file into an integrated data structure ready for counting read densities and testing differential occupancy.

Differential binding analysis: Date : 1/10/26

Here, we are looking into hoe many reads are suporting each peaks in all the samples,

creation of samplesheet.csv where we wil define input files and path:

(base) ibab@IBAB-MSc-Comp11:~/bio-env/mutant fastq files$ cat << 'EOF' > sample_sheet.csv
SampleID,Tissue,Factor,Condition,Replicate,bamReads,ControlID,bamControl,Peaks,PeakCaller
S1,yeast,ChIP,WT,1,/home/ibab/bio-env/mutant fastq
files/C1.sorted.bam,I1,/home/ibab/bio-env/mutant fastq
files/I1.sorted.bam,/home/ibab/bio-env/mutant fastq
files/WT_Rep1_narrow_peaks.narrowPeak,narrow
S2,yeast,ChIP,WT,2,/home/ibab/bio-env/mutant fastq
files/C2.sorted.bam,I2,/home/ibab/bio-env/mutant fastq
files/I2.sorted.bam,/home/ibab/bio-env/mutant fastq
files/WT_Rep2_narrow_peaks.narrowPeak,narrow
S3,yeast,ChIP,Mutant,1,/home/ibab/bio-env/mutant fastq
files/MC1.sorted.bam,MI1,/home/ibab/bio-env/mutant fastq files/MI1.sorted.bam,/home/ibab/bio-env/mutant fastq files/Mutant_Rep1_narrow_peaks.narrowPeak,narrow
S4,yeast,ChIP,Mutant,2,/home/ibab/bio-env/mutant fastq
files/MC2.sorted.bam,MI2,/home/ibab/bio-env/mutant fastq files/MI2.sorted.bam,/home/ibab/bio-env/mutant fastq files/Mutant_Rep2_narrow_peaks.narrowPeak,narrow
EOF



	A	B	C	D	E	F	G	H	I	J	K
1	SampleID	Tissue	Factor	Condition	Replicate	bamReads	ControlID	bamControl	Peaks	PeakCaller	
2	S1	yeast	ChIP	WT	1	/home/ibab/bio-env/mutant fastq files/C1.sorted.bam	I1	/home/ibab/bio-env/mutant fastq files/I1.sorted.bam	/home/ibab/bio-env/mutant fastq files/WT_Rep1_narrow_peaks.narrowPeak	narrow	
3	S2	yeast	ChIP	WT	2	/home/ibab/bio-env/mutant fastq files/C2.sorted.bam	I2	/home/ibab/bio-env/mutant fastq files/I2.sorted.bam	/home/ibab/bio-env/mutant fastq files/WT_Rep2_narrow_peaks.narrowPeak	narrow	
4	S3	yeast	ChIP	Mutant	1	/home/ibab/bio-env/mutant fastq files/MC1.sorted.bam	MI1	/home/ibab/bio-env/mutant fastq files/MI1.sorted.bam	/home/ibab/bio-env/mutant fastq files/Mutant_Rep1_narrow_peaks.narrowPeak	narrow	
5	S4	yeast	ChIP	Mutant	2	/home/ibab/bio-env/mutant fastq files/MC2.sorted.bam	MI2	/home/ibab/bio-env/mutant fastq files/MI2.sorted.bam	/home/ibab/bio-env/mutant fastq files/Mutant_Rep2_narrow_peaks.narrowPeak	narrow	
6											
7											

pixi run R

tam = dba(sampleSheet="sample_sheet.csv")

```
4 Samples, 1122 sites in matrix (2848 total):
  ID Tissue Factor Condition Replicate Intervals
1 S1  yeast   ChIP        WT         1      1188
2 S2  yeast   ChIP        WT         2      1818
3 S3  yeast   ChIP    Mutant         1      1412
4 S4  yeast   ChIP    Mutant         2        98
```

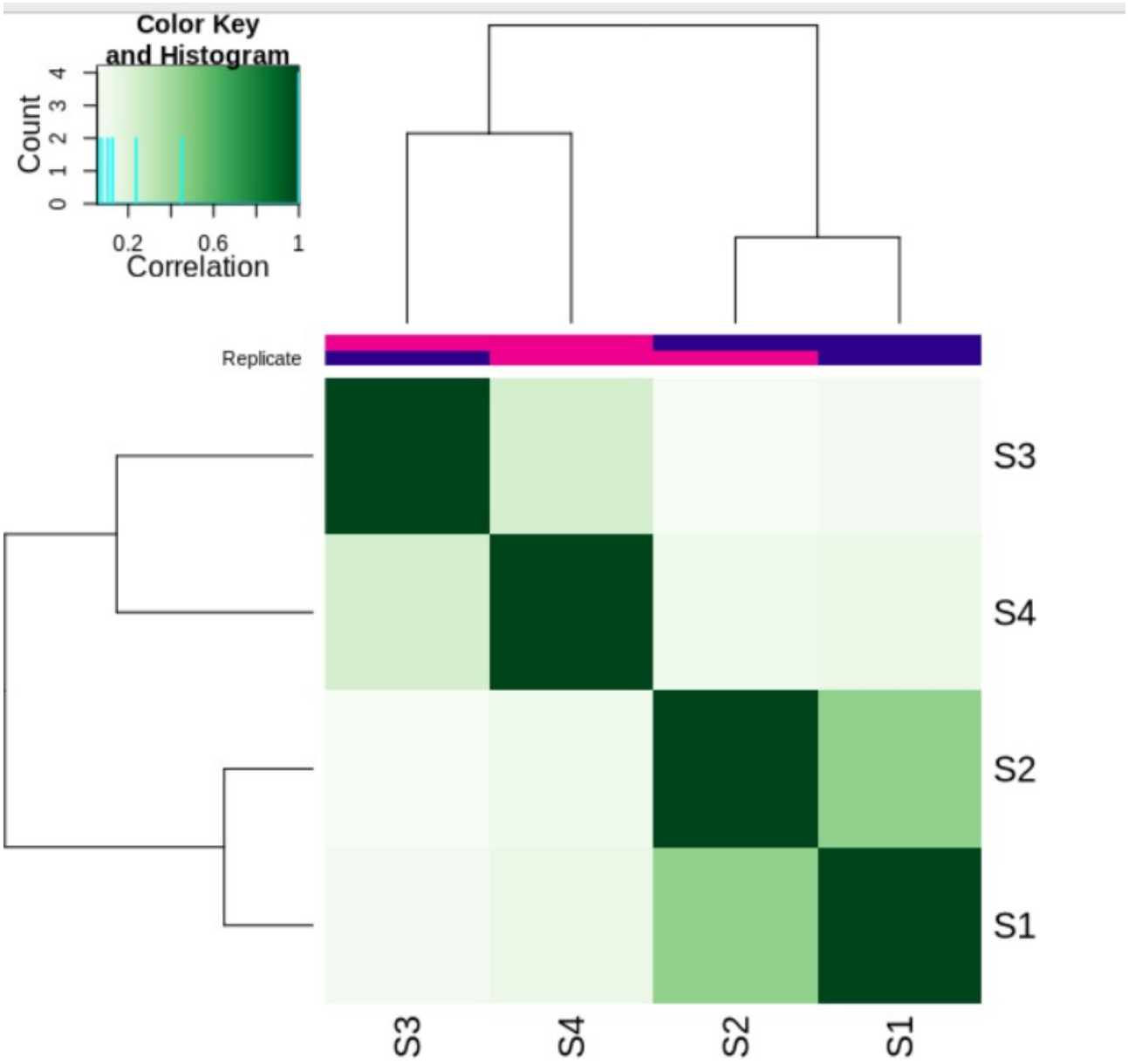
```
 tam <-  dba.count(tam)
```

```
4 Samples, 1110 sites in matrix:
  ID Tissue Factor Condition Replicate   Reads       FRiP
1 S1  yeast   ChIP        WT         1         3147941      0.08
2 S2  yeast   ChIP        WT         2         3520038      0.08
3 S3  yeast   ChIP    Mutant         1         776190       0.09
4 S4  yeast   ChIP    Mutant         2         434654       0.08
```

```
info <- dba.show(tam)
libsizes <- cbind(
  LibReads  = info$Reads,
  FRiP      = info$FRiP,
  PeakReads = round(info$Reads * info$FRiP)
)
rownames(libsizes) <- info$ID
libsizes
```

```
libsizes
   LibReads FRiP PeakReads
S1  3147941 0.08    251835
S2  3520038 0.08    281603
S3   776190 0.09     69857
S4   434654 0.08     34772
```

```
plot(tam)
```



1. Normalize the count data (default: sequencing depth)
tam <- dba.normalize(tam)

2. Retrieve normalization details
norm <- dba.normalize(tam, bRetrieve = TRUE)
norm

#####
$norm.method
[1] "lib"

$norm.factors
[1] 1.5981782 1.7870884 0.3940639 0.2206695

```
$lib.method
[1] "full"

$lib.sizes
[1] 3147941 3520038  776190  434654

$control.subtract
[1] TRUE

>
```

```
# 3. View table of raw library sizes vs. normalized library sizes
info <- dba.show(tam)
normlibs <- cbind(
  FullLibSize = norm$lib.sizes,
  NormFacs    = round(norm$norm.factors, 4),
  NormLibSize = round(norm$lib.sizes / norm$norm.factors)
)
rownames(normlibs) <- info$ID
normlibs
```

	FullLibSize	NormFacs	NormLibSize
S1	3147941	1.5982	1969706
S2	3520038	1.7871	1969706
S3	776190	0.3941	1969706
S4	434654	0.2207	1969706

Establishing contrast:

```
tam <- dba.contrast(
  tam,
  categories = DBA_CONDITION,
  minMembers = 2
)
tam
```

```
tam
Clearing existing contrasts.
4 Samples, 1110 sites in matrix:
  ID Tissue Factor Condition Replicate   Reads FRiP
1 S1  yeast   ChIP        WT         1 3147941 0.08
2 S2  yeast   ChIP        WT         2 3520038 0.08
3 S3  yeast   ChIP    Mutant         1  776190 0.09
4 S4  yeast   ChIP    Mutant         2  434654 0.08

Design: [~Condition] | 1 Contrast:
     Factor  Group Samples Group2 Samples2
1 Condition Mutant       2     WT        2
>
```

```
# ------------------------------------------------------------
# 5. Differential binding analysis
# ------------------------------------------------------------

tam_analyzed <- dba.analyze(
  tam,
  method = DBA_DESEQ2,
  bParallel = FALSE
)

# Check the results summary
dba.show(tam_analyzed, bContrasts = TRUE)
```

```
final dispersion estimates
    Factor  Group Samples Group2 Samples2 DB.DESeq2
1 Condition Mutant       2     WT        2        26
```

```
tam_analyzed$contrast
```

```
        id          pval          padj          fold
656    656 8.533690e-13 9.472396e-10 -1.102757e+01
165    165 1.602078e-10 8.891534e-08  3.641683e+00
984    984 3.719315e-09 1.376146e-06  8.997517e+00
56      56 1.123574e-07 3.117917e-05  8.616170e+00
663    663 1.992469e-07 4.423281e-05  8.478614e+00
882    882 3.284121e-07 6.075624e-05 -7.455111e+00
329    329 4.819582e-07 7.642481e-05  1.644940e+00
73      73 1.048996e-06 1.455482e-04 -7.431359e+00
223    223 3.814747e-06 4.704855e-04  1.589538e+00
567    567 4.405986e-06 4.890644e-04  7.762615e+00
```

```
# ------------------------------------------------------------
```

```
# 6. Export all regions (significant + non-significant)
# ------------------------------------------------------------

# 1. Ensure the output directory exists
if (!dir.exists("DiffBind_results")) {
  dir.create("DiffBind_results")
}

# 2. Extract all regions with normalized sample counts
res <- dba.report(
  tam_analyzed,
  th = 1,
  bCounts = TRUE
)

# 3. Convert GRanges to standard data frame
res_df <- as.data.frame(res)

# 4. Save to CSV
write.csv(
  res_df,
  "DiffBind_results/DiffBind_all_regions.csv",
  row.names = FALSE
)


# ------------------------------------------------------------
# 7. Significant regions (FDR < 0.05)
# ------------------------------------------------------------

res_sig <- res_df[
  !is.na(res_df$FDR) &
  res_df$FDR < 0.05,
]

write.csv(
  res_sig,
  "DiffBind_results/DiffBind_significant_regions_FDR0.05.csv",
  row.names = FALSE
)

# Print a quick breakdown of results
cat("Total analyzed regions: ", nrow(res_df), "\n")
cat("Significant regions (FDR < 0.05): ", nrow(res_sig), "\n")
cat("  - Enriched in Mutant (Fold > 0): ", sum(res_sig$Fold > 0), "\n")
cat("  - Enriched in WT (Fold < 0):     ", sum(res_sig$Fold < 0), "\n")
```

```
Total analyzed regions:  1110
Significant regions (FDR < 0.05):  26
  - Enriched in Mutant (Fold > 0):  22
- Enriched in WT (Fold < 0):      4
```

```
# 8. Classify significant regions
# ------------------------------

res_sig$Direction <- ifelse(
    res_sig$Fold > 0,
    "dGcn4_enriched",
    "WT_enriched"
)

write.csv(
    res_sig,
    "DiffBind_results/DiffBind_
    row.names=FALSE
)




# ------------------------------------------------------------
# 9. Summary & Save Object
# ------------------------------------------------------------

cat("\n=====================================\n")
cat("DIFFERENTIAL BINDING SUMMARY\n")
cat("=====================================\n")

cat("Total regions:", nrow(res_df), "\n")

cat(
  "Significant regions FDR < 0.05:",
  nrow(res_sig),
  "\n"
)

cat(
  "dGcn4-enriched:",
  sum(res_sig$Direction == "dGcn4_enriched"),
  "\n"
)

cat(
  "WT-enriched:",
  sum(res_sig$Direction == "WT_enriched"),
  "\n"
)

# Save the analyzed DiffBind object to disk
saveRDS(
  tam_analyzed,
```

```
  "DiffBind_results/DiffBind_final.rds"
)
```

```
====================================
DIFFERENTIAL BINDING SUMMARY
====================================
Total regions: 1110
Significant regions FDR < 0.05: 26
dGcn4-enriched: 22
WT-enriched: 4
```

Peak length distributions::

```
library(ggplot2)
library(dplyr)

# 1. Define paths to your narrowPeak files
peak_files <- list(
  "WT_Rep1"     = "WT_Rep1_narrow_peaks.narrowPeak",
  "WT_Rep2"     = "WT_Rep2_narrow_peaks.narrowPeak",
  "Mutant_Rep1" = "Mutant_Rep1_narrow_peaks.narrowPeak",
  "Mutant_Rep2" = "Mutant_Rep2_narrow_peaks.narrowPeak"
)

# 2. Function to read a narrowPeak file and calculate length
read_peak_length <- function(filepath, sample_name) {
  df <- read.table(filepath, header = FALSE, sep = "\t", stringsAsFactors = FALSE)
  # Col 2: chromStart, Col 3: chromEnd
  data.frame(
    Sample    = sample_name,
    Condition = ifelse(grepl("WT", sample_name), "WT", "Mutant"),
    Length    = df$V3 - df$V2
  )
}

# 3. Combine into a single tidy data frame
all_peak_lengths <- bind_rows(
  lapply(names(peak_files), function(nm) {
    read_peak_length(peak_files[[nm]], nm)
  })
)

# Inspect summary metrics per sample
all_peak_lengths %>%
  group_by(Sample) %>%
  summarise(
    Peak_Count = n(),
```

```
  Min_Length = min(Length),
  Median_Length = median(Length),
  Mean_Length = round(mean(Length), 1),
  Max_Length = max(Length)
)
```

```
# A tibble: 4 × 6
  Sample      Peak_Count Min_Length Median_Length Mean_Length Max_Length
  <chr>            <int>      <int>         <dbl>       <dbl>      <int>
1 Mutant_Rep1       1412        122           249        283.        997
2 Mutant_Rep2         98        109           188        231.        989
3 WT_Rep1           1188        140           248        320.       2356
4 WT_Rep2           1818        139           257        341.       2299
> 
```

ALL differential regions:

```
cat << 'EOF' > export_results.R
library(DiffBind)
library(BiocParallel)

# 1. Enforce single-core processing to prevent RAM spikes
register(SerialParam())

# 2. Ensure output directory exists
output_dir <- "DiffBind_results"
if (!dir.exists(output_dir)) {
  dir.create(output_dir)
}

cat("\n[1/5] Loading sample sheet...\n")
tam <- dba(sampleSheet = "sample_sheet.csv")

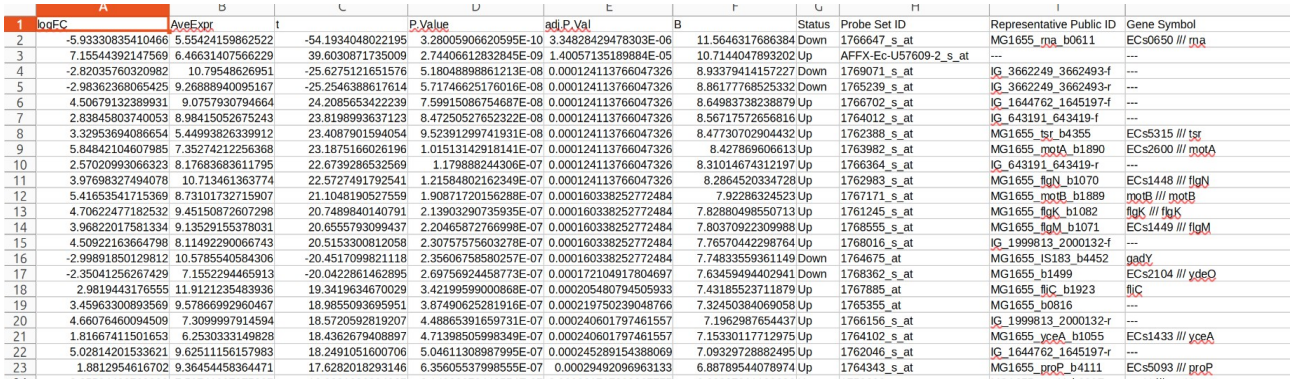
cat("[2/5] Counting reads safely...\n")
tam <- dba.count(tam, bParallel = FALSE)

cat("[3/5] Normalizing data and establishing contrast...\n")
tam <- dba.normalize(tam)
tam <- dba.contrast(tam, categories = DBA_CONDITION, minMembers = 2)

cat("[4/5] Running differential binding analysis...\n")
tam_analyzed <- dba.analyze(tam, method = DBA_DESEQ2, bParallel = FALSE)

# Save the analyzed DBA object to disk
EOF("Files saved to:", output_dir, "\n\n")n")es_sig$Fold < 0), "\n")ns_FDR0.05.csv"), row.names =
FALSE)
```

```
==================================
DIFFERENTIAL BINDING ANALYSIS COMPLETE
==================================
Total analyzed regions  :  1108
Significant (FDR < 0.05):  27
- Mutant Enriched (Fold > 0):  23
- WT Enriched     (Fold < 0):  4
==================================
Files saved to: DiffBind_results
```



	A	B	C	D	E	F	G	H	I	
1	logFC	AveExpr	t	P.Value	adj.P.Val	B	Status	Probe Set ID	Representative Public ID	Gene Symbol
2	-5.93330835410466	5.55424159862522	-54.1934048022195	3.28005906620595E-10	3.34828429478303E-06	11.5646317686384	Down	1766647_s_at	MG1655_rna_b0611	ECs0650 /// rna
3	7.15544392147569	6.46631407566229	39.6030871735009	2.74406612832845E-09	1.40057135189884E-05	10.7144047893202	Up	AFFX-Ec-U57609-2_s_at	---	---
4	-2.82035760320982	10.79548626951	-25.6275121651576	5.18048898861213E-08	0.000124113766047326	8.93379414157227	Down	1769071_s_at	IG_3662249_3662493-f	---
5	-2.98362368065425	9.26888940095167	-25.2546388617614	5.71746625176016E-08	0.000124113766047326	8.86177768525332	Down	1765239_s_at	IG_3662249_3662493-r	---
6	4.50679132389931	9.07579307946664	24.2085653422239	7.59915086754687E-08	0.000124113766047326	8.64983738238879	Up	1766702_s_at	IG_1644762_1645197-f	---
7	2.83845803740053	8.98415052675243	23.8198993637123	8.47250527652322E-08	0.000124113766047326	8.56717572656816	Up	1764012_s_at	IG_643191_643419-f	---
8	3.32953694086654	5.44993826339912	23.4087901594054	9.52391299741931E-08	0.000124113766047326	8.47730702904432	Up	1762388_s_at	MG1655_tsr_b4355	ECs5315 /// tsr
9	5.84842104607985	7.35274212256368	23.1875166026196	1.01513142918141E-07	0.000124113766047326	8.427869606613	Up	1763982_s_at	MG1655_motA_b1890	ECs2600 /// motA
10	2.57020993066323	8.17683683611795	22.6739286532569	1.179888244306E-07	0.000124113766047326	8.31014674312197	Up	1766364_s_at	IG_643191_643419-r	---
11	3.97698327494078	10.713461363774	22.5727491792541	1.21584802162349E-07	0.000124113766047326	8.2864520334728	Up	1762983_s_at	MG1655_flgN_b1070	ECs1448 /// flgN
12	5.41653541715369	8.73101732715907	21.1048190527559	1.90871720156288E-07	0.000160338252772484	7.92286324523	Up	1767171_s_at	MG1655_motB_b1889	motB /// motB
13	4.70622477182532	9.45150872607298	20.7489840140791	2.13903290735935E-07	0.000160338252772484	7.82880498550713	Up	1761245_s_at	MG1655_flgK_b1082	flgK /// flgK
14	3.96822017581334	9.13529155378031	20.6555793099437	2.20465872766998E-07	0.000160338252772484	7.80370922309988	Up	1768555_s_at	MG1655_flgM_b1071	ECs1449 /// flgM
15	4.50922163664798	8.11492290066743	20.5153300812058	2.30757575603278E-07	0.000160338252772484	7.76570442298764	Up	1768016_s_at	IG_1999813_2000132-f	---
16	-2.99891850129812	10.5785540584306	-20.4517099821118	2.35606758580257E-07	0.000160338252772484	7.74833559361149	Down	1764675_at	MG1655_IS183_b4452	gadY
17	-2.35041256267429	7.1552294465913	-20.0422861462895	2.69756924458773E-07	0.000172104917804697	7.63459494402941	Down	1768362_s_at	MG1655_b1499	ECs2104 /// ydeO
18	2.98194431765555	11.9121235483936	19.3419634670029	3.42199599000868E-07	0.000205480794505933	7.43185523711879	Up	1767885_at	MG1655_fliC_b1923	fliC
19	3.45963300893569	9.57866992960467	18.9855093695951	3.87490625281916E-07	0.000219750239048766	7.32450384069058	Up	1765355_at	MG1655_b0816	---
20	4.66076460094509	7.3099997914594	18.5720592819207	4.48865391659731E-07	0.000240601797461557	7.1962987654437	Up	1766156_s_at	IG_1999813_2000132-r	---
21	1.81667411501653	6.2530333149828	18.4362679408897	4.71398505998349E-07	0.000240601797461557	7.15330117712975	Up	1764102_s_at	MG1655_yceA_b1055	ECs1433 /// yceA
22	5.02814201533621	9.62511156157983	18.2491051600706	5.04611308987995E-07	0.000245289154388069	7.09329728882495	Up	1762046_s_at	IG_1644762_1645197-r	---
23	1.8812954616702	9.36454458364471	17.6282018293146	6.35605537998555E-07	0.000294920969633	6.88789544078974	Up	1764343_s_at	MG1655_proP_b4111	ECs5093 /// proP

Downloading the gff file

wget -O saccharomyces_cerevisiae_S288C.gff.gz "https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/146/045/GCF_000146045.2_R64/GCF_000146045.2_R64_genomic.gff.gz"

1. Check file size
ls -lh saccharomyces_cerevisiae_S288C.gff.gz

2. Decompress GFF file
gunzip saccharomyces_cerevisiae_S288C.gff.gz

3. Extract unique chromosome/sequence identifiers
grep -v '^#' saccharomyces_cerevisiae_S288C.gff | cut -f1 | sort -u | head -15

Conversion of Significant Regions from CSV to BED Format:

```
# 1. Read the CSV file into R
res_sig <- read.csv("DiffBind_results/DiffBind_significant_regions_FDR0.05.csv")

# 2. Export coordinates to BED format
write.table(
  res_sig[, c("seqnames", "start", "end")],
  file = "DiffBind_significant_regions_FDR0.05.bed",
  sep = "\t",
  quote = FALSE,
  row.names = FALSE,
  col.names = FALSE
)

cat("BED file successfully generated!\n")
```

```
> res_sig
       seqnames   start     end width strand      Conc Conc_Mutant   Conc_WT
1   NC_001144.5  460150  460550   401      * 10.226136    0.000000 11.226136
2  NC_001136.10  576250  576650   401      *  8.279173    9.180699  5.357344
3   NC_001148.4   39183   39583   401      *  5.496164    6.496164  0.000000
4   NC_001134.8  544132  544532   401      *  5.212116    6.212116  0.000000
5   NC_001144.5  501154  501554   401      *  5.092670    6.092670  0.000000
6   NC_001146.8  783983  784383   401      *  6.363140    0.000000  7.363140
7   NC_001138.5  223451  223851   401      *  7.041224    7.749878  5.590041
8   NC_001135.5      39     439   401      *  6.462122    0.000000  7.462122
9  NC_001136.10 1277613 1278013   401      *  6.357090    7.073384  4.871265
10  NC_001143.9  201308  201708   401      *  4.688448    5.688448  0.000000
11  NC_001144.5  490177  490577   401      *  5.042156    6.036018  0.000000
12  NC_001143.9  613543  613943   401      *  6.555578    7.318348  4.834179
13  NC_001143.9  416274  416674   401      *  5.798728    6.602248  3.825287
14  NC_001144.5  131521  131921   401      *  5.280859    6.136072  2.892303
15  NC_001224.1   33000   33400   401      *  7.555789    8.212568  6.316022
16  NC_001142.9  542386  542786   401      *  5.369725    6.165521  3.448141
17  NC_001135.5   57015   57415   401      *  7.055999    7.847921  5.159642
18  NC_001139.9   63036   63436   401      *  5.707411    6.421857  4.230055
19  NC_001146.8  621898  622298   401      *  4.806691    5.730918  1.518027
20  NC_001141.2  159860  160260   401      *  5.988640    6.619643  4.840987
21  NC_001142.9  554301  554701   401      *  5.617416    6.551577  2.130927
22  NC_001141.2  187060  187460   401      *  6.205115    6.852444  4.999998
23  NC_001146.8   90242   90642   401      *  5.925632    6.566830  4.742440
24 NC_001136.10  562013  562413   401      *  6.365911    7.208588  4.091000
```

HOMER Genomic Region Annotation:

```
annotatePeaks.pl DiffBind_significant_regions_FDR0.05.bed reference.fna -gff3
saccharomyces_cerevisiae_S288C.gff > DiffBind_significant_regions_HOMER_annotated.txt
```

HOMER Motif analysis:

```
findMotifsGenome.pl DiffBind_significant_regions_FDR0.05.bed reference.fna motif_analysis
-size 200
```

Output:

File / Folder	Meaning
homerResults.html	**Main summary report** of discovered motifs. Open this in a web browser.
knownResults.html	Summary of **known DNA motifs** found in your regions.
homerResults/	Detailed results for **de novo motifs** discovered by HOMER.
knownResults/	Detailed results for **known motifs** matching your sequences.
homerMotifs.all.motifs	All motifs identified by HOMER.
homerMotifs8, homerMotifs10, homerMotifs12	Motif files for different motif lengths.
knownResults.txt	Text version of known-motif results.
motifFindingParameters.txt	Parameters/settings used for the motif analysis.
seq.autonorm.tsv	Sequence normalization information used by HOMER.
nonRedundant.motifs	Non-redundant set of identified motifs.

```
12 of 25 Initial Sequence: TAATAATAATAA... (-8.549)
        Round 1: -8.55 TAATAATAATAA T:18.0(49.30%),B:7022.2(17.10%),P:1e-3
        Round 2: -8.55 TAATAATAATAA T:18.0(49.30%),B:7022.2(17.10%),P:1e-3
        =Final=: -3.05 TAATAATAATAA T:5.0(18.52%),B:2821.3(7.47%),P:1e-1
        Performing exhaustive masking of motif...
        Reprioritizing potential motifs...
Remaining seeds don't look promising (After initial 5 motifs, logp -7.235 > -7.431)

Finalizing Enrichment Statistics (new in v3.4)
Reading input files...
56517 total sequences read
Cache length = 11180
Using binomial scoring
Checking enrichment of 12 motif(s)
|0%                                  50%                                  100%|
================================================================================
Output in file: motif_analysis/homerMotifs.motifs12
```

Rank	Motif	P-value	log P-pvalue	% of Targets	% of Background	STD(Bg STD)	Best Match/Details	Motif File
1 *		1e-8	-1.854e+01	20.00%	0.03%	23.0bp (34.2bp)	PB0109.1_Bbx_2/Jaspar(0.742) More Information | Similar Motifs Found	motif file (matrix)
2 *		1e-5	-1.179e+01	13.33%	0.03%	28.0bp (30.9bp)	sage/MA2268.1/Jaspar(0.731) More Information | Similar Motifs Found	motif file (matrix)
3 *		1e-4	-9.848e+00	26.67%	1.45%	1.7bp (235.9bp)	MAZ/MA1522.2/Jaspar(0.647) More Information | Similar Motifs Found	motif file (matrix)
4 *		1e-4	-9.452e+00	20.00%	0.57%	34.6bp (46.3bp)	wc-1/MA1437.2/Jaspar(0.848) More Information | Similar Motifs Found	motif file (matrix)
5 *		1e-3	-7.456e+00	6.67%	0.00%	0.0bp (0.8bp)	Eip74EF/MA0026.1/Jaspar(0.796) More Information | Similar Motifs Found	motif file (matrix)
6 *		1e-3	-7.456e+00	6.67%	0.01%	0.0bp (30.5bp)	B52(RRM)/Drosophila_melanogaster-RNCMPT00134-PBM/HughesRNA(0.709) More Information | Similar Motifs Found	motif file (matrix)
7 *		1e-3	-7.456e+00	6.67%	0.01%	0.0bp (30.2bp)	NAC035/MA2046.2/Jaspar(0.710) More Information | Similar Motifs Found	motif file (matrix)